



LUND
UNIVERSITY

Written Examination
Linear Analysis, MATB24
November 17, 2021
Time: 13.00–18.00

Centre for Mathematical Sciences
Mathematics, Faculty of Science

You may freely use formulas from the formula sheet. Refer to relevant theorems from the course when you use them. Write clearly with concise motivations and explain your steps. In each of the six problems a maximum of five points can be achieved. No aids other than writing utensils are allowed.

1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(t) = |\sin t|$.

- a) Compute the real Fourier series of f and prove that it converges uniformly to f .
- b) Determine the value of

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}.$$

Solution:

- a) Obviously, f is even. We compute

$$\begin{aligned} \int_0^{\pi} |\sin t| dt &= \int_0^{\pi} \sin t dt = 2, \\ \int_0^{\pi} |\sin t| \cos t dt &= \int_0^{\pi} \sin t \cos t dt = \frac{1}{2} \sin^2 t \Big|_{t=0}^{\pi} = 0, \end{aligned}$$

and for $n \in \mathbb{N}$, $n \geq 2$,

$$\begin{aligned} \int_0^{\pi} \sin t \cos(nt) dt &= -\frac{1}{n} \int_0^{\pi} \cos t \sin(nt) dt \\ &= \frac{1}{n^2} \cos t \cos(nt) \Big|_{t=0}^{\pi} + \frac{1}{n^2} \int_0^{\pi} \sin t \cos(nt) dt, \end{aligned}$$

so

$$\int_0^{\pi} |\sin t| \cos(nt) dt = \int_0^{\pi} \sin t \cos(nt) dt = \frac{\frac{1}{n^2}((-1)^{n+1} - 1)}{1 - \frac{1}{n^2}} = \begin{cases} 0, & n \text{ odd}, \\ \frac{2}{1-n^2}, & n \text{ even}. \end{cases}$$

Thus,

$$f(t) \sim \frac{2}{\pi} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{\cos(2kt)}{4k^2 - 1}.$$

Since the coefficients are absolutely summable and f is continuous, the Fourier series converges uniformly to f .

- b) By a) we have

$$0 = f(0) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{4k^2 - 1},$$

so

$$\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2}.$$

Please, turn over!

2. Find a solution $u = u(x, t)$ for $(x, t) \in [0, \pi/2] \times [0, \infty)$ to the following problem for the heat equation:

$$\begin{cases} 3u_t = u_{xx}, & 0 < x < \pi/2, t > 0, \\ u_x(0, t) = u(\pi/2, t) = 0, & t > 0, \\ u(x, 0) = \cos^3(3x), & 0 < x < \pi/2. \end{cases}$$

Also briefly discuss the regularity of your solution and in which sense the initial condition is satisfied for it.

Solution: We follow the method of separation of variables and begin by inserting the ansatz $u(x, t) = X(x)T(t)$ ($X \neq 0, T \neq 0$) into the heat equation. This yields

$$XT' = \frac{1}{3}X''T \Rightarrow X'' = \lambda X, T' = \frac{1}{3}\lambda T$$

for some constant $\lambda \in \mathbb{R}$. Equipping the equation for X with the boundary conditions $X'(0) = 0, X(\pi/2) = 0$, we can only get nontrivial solutions for $\lambda < 0$, namely

$$X(x) = a \sin(\sqrt{-\lambda}x) + b \cos(\sqrt{-\lambda}x)$$

for some constants $a, b \in \mathbb{R}$. Due to the boundary condition $X'(0) = 0$, necessarily $a = 0$, and by the other boundary condition $X'(\pi/2) = 0$ we obtain a nontrivial solution if and only if

$$\cos(\sqrt{-\lambda}\pi/2) = 0 \Leftrightarrow \lambda = -(2n+1)^2, n \in \mathbb{N}_0.$$

To sum up, we obtain $X_n(x) = b_n \cos((2n+1)x)$ for some constant $b_n \in \mathbb{R}$ and where $n \in \mathbb{N}_0$. On the other hand, the Cauchy problem $T'_n = -\frac{1}{3}(2n+1)^2 T_n, T_n(0) = 1$ (say) has the unique solution $T_n(t) = e^{-\frac{1}{3}(2n+1)^2 t}$, where $n \in \mathbb{N}_0$. Therefore, we get the solutions $u_n(x, t) = b_n e^{-\frac{1}{3}(2n+1)^2 t} \cos((2n+1)x)$ (of the heat equation together with the boundary conditions), $n \in \mathbb{N}_0$, and superposing them yields (formally) the trial solution

$$u(x, t) = \sum_{n=0}^{\infty} b_n e^{-\frac{1}{3}(2n+1)^2 t} \cos((2n+1)x).$$

Moreover,

$$\cos^3(3x) = \left(\frac{e^{3ix} + e^{-3ix}}{2} \right)^3 = \frac{e^{9ix} + e^{-9ix}}{8} + \frac{3(e^{3ix} + e^{-3ix})}{8} = \frac{1}{4} \cos(9x) + \frac{3}{4} \cos(3x)$$

for $x \in \mathbb{R}$. Therefore, choose $b_1 := 3/4, b_4 := 1/4$, and $b_n := 0$ for $n \in \mathbb{N}_0 \setminus \{1, 4\}$; thus,

$$u(x, t) = \frac{3}{4} e^{-3t} \cos(3x) + \frac{1}{4} e^{-27t} \cos(9x).$$

Clearly, $u \in C^\infty(\mathbb{R}^2)$ and $u(x, 0) = \cos^3(3x)$ for $0 < x < \pi/2$.

3. a) Prove that (K_n) , given by

$$K_n: \mathbb{R} \rightarrow \mathbb{R}, \quad K_n(x) = \frac{n}{\pi(1 + n^2x^2)}, \quad n \in \mathbb{N},$$

is a positive summation kernel on $\mathbb{R}$.

- b) Compute

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} \frac{ne^{-x^2}}{1 + n^2x^2} dx.$$

Solution:

- a) We check the three properties:

- (i) Clearly, $K_n \geq 0$ for all $n \in \mathbb{N}$.
(ii) For all $n \in \mathbb{N}$:

$$\begin{aligned} \int_{-\infty}^{\infty} K_n(x) dx &\stackrel{y=nx}{=} \int_{-\infty}^{\infty} \frac{1}{\pi(1 + y^2)} dy = \lim_{r \rightarrow \infty} \frac{1}{\pi} \arctan y \Big|_{y=-r}^r = \frac{1}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) \\ &= 1. \end{aligned}$$

- (iii) For all $\delta > 0$:

$$\int_{|x| > \delta} K_n(x) dx \stackrel{y=nx}{=} \int_{|y| > n\delta} \frac{1}{\pi(1 + y^2)} dy \xrightarrow{n \rightarrow \infty} 0.$$

- b) Let $f(x) := e^{-x^2}$, $x \in \mathbb{R}$. Since f is integrable, continuous, and bounded, it follows from a) that

$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} \frac{ne^{-x^2}}{1 + n^2x^2} dx = \pi \lim_{n \rightarrow \infty} (K_n * f)(0) = \pi f(0) = \pi.$$

Please, turn over!

4. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$,

$$f(t) = e^{-3|t|}, \quad g(t) = \begin{cases} |t|, & |t| \leq 1, \\ 0, & |t| > 1. \end{cases}$$

- a) Compute the Fourier transforms of f and g .
b) Determine whether or not a function $h \in L^1(\mathbb{R})$ exists such that

$$\int_{-\infty}^{\infty} h(t-s)f(s) ds = g(t)$$

for all $t \in \mathbb{R}$, and, if yes, provide an example of such an h .

Solution:

- a) We compute

$$\hat{f}(\omega) \stackrel{f \text{ even}}{=} 2 \operatorname{Re} \int_0^{\infty} e^{(-3-i\omega)t} dt = 2 \operatorname{Re} \frac{1}{3+i\omega} = \frac{6}{9+\omega^2}, \quad \omega \in \mathbb{R},$$

and

$$\begin{aligned} \hat{g}(0) &= 2 \int_0^1 t dt = 1, \\ \hat{g}(\omega) &\stackrel{g \text{ even}}{=} 2 \int_0^1 t \cos(\omega t) dt = \frac{2 \sin \omega}{\omega} - \frac{2}{\omega} \int_0^1 \sin(\omega t) dt \\ &= 2 \left(\frac{\sin \omega}{\omega} + \frac{\cos \omega - 1}{\omega^2} \right), \quad \omega \in \mathbb{R} \setminus \{0\}. \end{aligned}$$

- b) Suppose that such an h exists. Then $\hat{g} = \widehat{h * f} = \hat{h} \hat{f}$, hence

$$\hat{h}(\omega) = \frac{\hat{g}(\omega)}{\hat{f}(\omega)} = \frac{\omega \sin \omega}{3} + \frac{3 \sin \omega}{\omega} + \frac{(\cos \omega - 1)(9 + \omega^2)}{3\omega^2}, \quad \omega \in \mathbb{R} \setminus \{0\}.$$

Obviously, $\hat{h}$ is not bounded (last two terms bounded, but first not), which, however, cannot happen provided $h \in L^1(\mathbb{R})$. Thus, such an h does not exist.

5. a) Prove that the series

$$\sum_{n=1}^{\infty} \frac{x^2}{n(1+nx^4)}$$

converges uniformly on $\mathbb{R}$.

- b) Compute

$$\lim_{k \rightarrow \infty} \sum_{n=1}^k \int_0^4 \frac{x^2 \cos(kx)}{n(1+nx^4)} dx.$$

Solution:

- a) Because of $1+nx^4 \geq 2\sqrt{nx^2}$ we have

$$\left| \frac{x^2}{n(1+nx^4)} \right| \leq \frac{1}{2n^{3/2}}$$

for all $x \in \mathbb{R}$, $n \in \mathbb{N}$. Weierstrass M -test yields the claim (notice $3/2 > 1$).

- b) Let us denote

$$f_k(x) := \sum_{n=1}^k \frac{x^2}{n(1+nx^4)}, \quad f(x) := \sum_{n=1}^{\infty} \frac{x^2}{n(1+nx^4)}, \quad k \in \mathbb{N}, x \in \mathbb{R}.$$

Now

$$\begin{aligned} \left| \sum_{n=1}^k \int_0^4 \frac{x^2 \cos(kx)}{n(1+nx^4)} dx \right| &= \left| \int_0^4 f_k(x) \cos(kx) dx \right| \\ &\leq \int_0^4 |f_k(x) - f(x)| dx + \left| \int_0^4 f(x) \cos(kx) dx \right|. \end{aligned}$$

Here, the first summand tends to zero as $k \rightarrow \infty$ by uniform convergence of f_k to f (due to a)) and the second also to zero by the Riemann–Lebesgue lemma (notice that f is continuous by a), so in particular absolutely integrable on $[0, 4]$). Therefore,

$$\lim_{k \rightarrow \infty} \sum_{n=1}^k \int_0^4 \frac{x^2 \cos(kx)}{n(1+nx^4)} dx = 0.$$

Please, turn over!

6. Decide whether or not a function with the respective properties exists. If yes, provide an example; if not, prove that no such function can exist.

a) $f \in C(\mathbb{T})$ with $\hat{f}(n) = \frac{1}{1+n^2}$ for all $n \in \mathbb{Z}$.

b) $f \in L^1(\mathbb{R})$ with

$$\hat{f}(\omega) = \begin{cases} \frac{1}{1+\omega^2}, & \omega > 0, \\ 0, & \omega \leq 0. \end{cases}$$

c) $f \in L^1(\mathbb{R})$ differentiable, bounded with $f \neq 0$ and $f * f = f$.

Solution:

a) Yes,

$$f(t) := \sum_{n \in \mathbb{Z}} \frac{e^{int}}{1+n^2}.$$

Indeed, by Weierstrass M -test this series converges uniformly and f is continuous.

b) No, necessarily $\hat{f}$ is continuous for $f \in L^1(\mathbb{R})$.

c) No. Let $f \in L^1(\mathbb{R})$ be differentiable, bounded with $f * f = f$. Then $(\hat{f})^2 = \widehat{f * f} = \hat{f}$, so $\hat{f}(\omega) \in \{0, 1\}$ for all $\omega \in \mathbb{R}$. By continuity of $\hat{f}$ and $\lim_{|\omega| \rightarrow \infty} \hat{f}(\omega) = 0$ it follows that $\hat{f} \equiv 0$ and thus $f \equiv 0$.